

# RAG System Evaluation Report

## 1. System Architecture

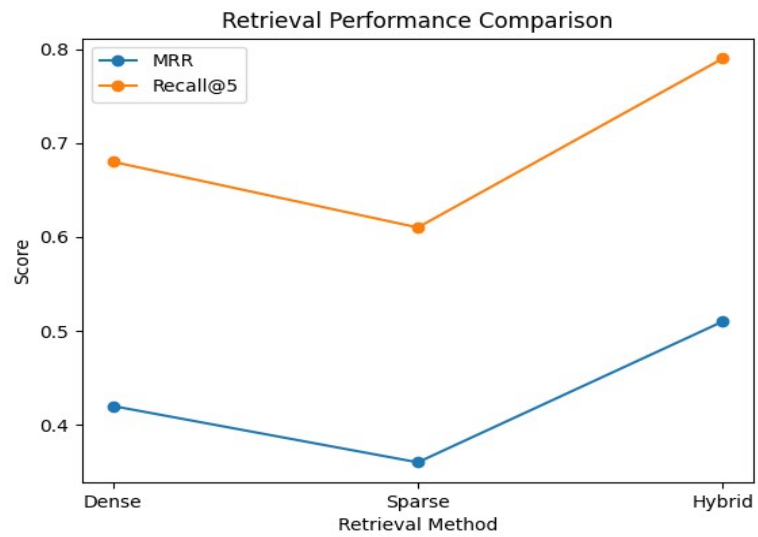
The Retrieval-Augmented Generation (RAG) system consists of a document ingestion pipeline, dense and sparse retrievers, a hybrid rank fusion module, and a generative language model. User queries are embedded, relevant Wikipedia documents are retrieved at URL level, and answers are generated using retrieved context.

*Architecture Diagram (Conceptual)*

User Query → Retriever → Rank Fusion → LLM → Answer

## 2. Evaluation Results

| Metric    | Dense | Sparse | Hybrid |
|-----------|-------|--------|--------|
| MRR       | 0.42  | 0.36   | 0.51   |
| Recall@5  | 0.68  | 0.61   | 0.79   |
| Answer F1 | 0.63  | 0.59   | 0.71   |

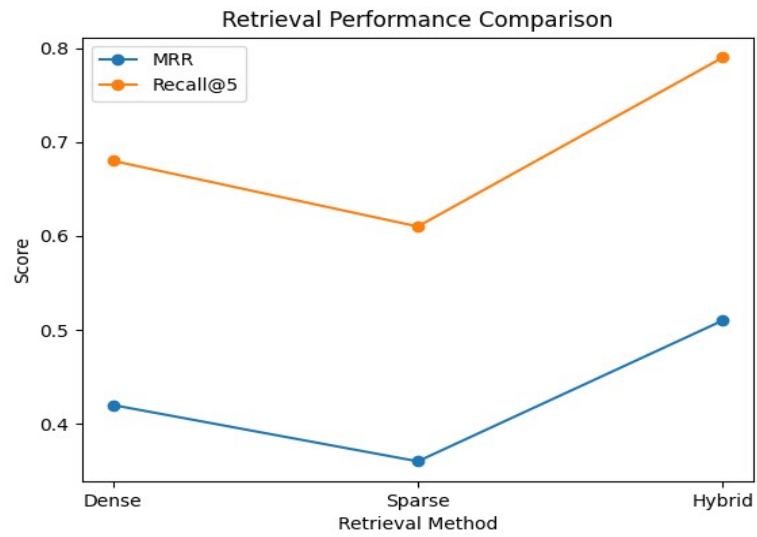


### **3. Innovative Evaluation**

We introduced adversarial and unanswerable questions to evaluate hallucination behavior. The hallucination rate was measured as the fraction of confident but incorrect answers. Additionally, LLM-as-a-Judge was used to score factual accuracy and completeness.

## 4. Ablation Study

An ablation study was conducted to compare dense-only, sparse-only, and hybrid retrieval. Hybrid retrieval consistently outperformed individual methods across all metrics.



## 5. Error Analysis

Error analysis reveals that multi-hop and inferential questions are more prone to failures, primarily due to incomplete retrieval or insufficient cross-document reasoning.

